

Project Requirement Document(PRD)

1. Executive Summary

Customer churn is one of the biggest challenges faced by businesses today.

Most companies only realize a customer has left after the loss has already happened.

Our platform helps businesses identify customers who are likely to leave, understand the reasons behind churn, estimate the financial impact, and recommend actions to retain valuable customers.

Instead of simply predicting churn, our solution helps businesses make smarter retention decisions.

2. Problem Statement

Businesses spend a significant amount of money acquiring customers.

However, many customers leave because businesses fail to identify warning signs early.

Current solutions often:

- Detect churn too late
- Do not explain why customers leave
- Do not estimate business impact
- Do not provide actionable recommendations

As a result, companies lose customers and revenue.

3. Proposed Solution

We are building an AI-powered platform that:

- 1 Predicts customer churn
- 2 Explains the reasons behind churn
- 3 Calculates customer risk levels
- 4 Estimates potential revenue loss
- 5 Recommends retention strategies
- 6 Provides insights through an interactive dashboard

The platform transforms raw customer data into business decisions.

4. Target Users

Primary Users

- Business Managers
- Customer Success Teams
- Marketing Teams
- CRM Teams
- Sales Teams

Secondary Users

- Data Analysts
- Business Analysts
- Executives

5. Product Goals

Our platform aims to:

- Reduce customer churn
- Improve customer retention
- Increase business revenue
- Support better decision making
- Prioritize high-value customers
- Provide actionable recommendations

6. Success Metrics

The project will be considered successful if it can:

- Predict churn accurately
- Identify major churn drivers
- Segment customers effectively
- Estimate revenue at risk
- Generate useful recommendations

Key KPIs:

- Churn Rate
- Customer Retention Rate
- Revenue at Risk
- High-Risk Customer Count
- Customer Health Score

7. Product Workflow

Customer Data



Data Cleaning

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Business Analysis

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Exploratory Data Analysis

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Machine Learning Model

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Churn Prediction

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Root Cause Analysis

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Customer Segmentation

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Revenue Impact Analysis

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Recommendation Engine

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Power BI Dashboard

8. Core Features

Feature 1: Business Analytics

Description

Analyze customer behavior using SQL and visualizations.

Insights Generated

- Churn by contract type
- Churn by payment method
- Churn by tenure
- Revenue trends

- Customer behavior patterns

Value

Helps businesses understand customer trends.

Feature 2: Churn Prediction Engine

Description

Uses machine learning to identify customers likely to leave.

Input

Customer information

Output

- Churn probability
- Risk level
- Prediction result

Example

Customer A

Churn Probability: 92%

Status: High Risk

Feature 3: Explainable AI

Description

Explains why a customer is likely to churn.

Output Example

Reasons:

- High monthly charges
- Short tenure
- No support services
- Month-to-month contract

Value

Makes predictions easier to understand and trust.

Feature 4: Customer Health Score

Description

Measures the overall relationship strength between customer and company.

Score Range

0–100

Categories

- Healthy
- Moderate Risk
- High Risk
- Critical

Value

Quickly identifies customers needing attention.

Feature 5: Customer Segmentation

Description

Groups customers based on risk and value.

Segments

- Loyal Customers
- Growth Customers
- High-Risk Customers
- Critical Customers

Value

Supports targeted retention strategies.

Feature 6: Revenue Impact Analysis

Description

Estimates the financial impact of churn.

Metrics

- Revenue at Risk
- Potential Revenue Loss
- Revenue Saved

Value

Connects AI predictions to business outcomes.

Feature 7: AI Recommendation Engine

Description

Suggests actions to reduce churn.

Examples

Customer Issue → Recommendation

High Charges → Offer Discount

Low Engagement → Personalized Offer

Frequent Complaints → Priority Support

New Customer → Onboarding Assistance

Value

Provides actionable business guidance.

Feature 8: Churn Playbook

Description

A decision framework for retention teams.

Example

High Risk

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High Revenue Customer

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Assign Relationship Manager

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Offer Retention Package

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Follow Up

Value

Converts predictions into actions.

Feature 9: Executive Dashboard

Description

Interactive Power BI dashboard for business users.

Dashboard KPIs

- Total Customers
- Churn Rate
- Revenue
- Revenue at Risk
- Customer Health Score
- High-Risk Customers

Visualizations

- Monthly Churn Trend
- Revenue Analysis
- Customer Segmentation
- Contract Analysis
- Risk Distribution

Value

Provides complete business visibility.

9. Technical Architecture

Frontend

- Power BI Dashboard
- Streamlit (Optional)

Backend

- Python

Analytics

- SQL
- Pandas
- NumPy

Machine Learning

- Scikit-Learn
- XGBoost

Explainable AI

- SHAP

Version Control

- GitHub

10. Data Requirements

Dataset Fields

- Customer ID
- Gender
- Senior Citizen
- Tenure
- Contract Type
- Internet Service
- Monthly Charges
- Total Charges
- Payment Method
- Churn Status

Source

Telco Customer Churn Dataset

11. Unique Selling Points (USP)

Most churn projects only answer:

"Who will leave?"

Our platform answers:

- Who will leave?
- Why will they leave?
- How much money is at risk?
- Which customers should be prioritized?
- What actions should be taken?

This makes our solution business-focused rather than prediction-focused.

12. Future Enhancements

Future versions may include:

- Real-time churn prediction
- CRM integration
- Automated email campaigns
- LLM-powered retention assistant
- Customer support integration

- Real-time customer monitoring

13. Project Deliverables

Final Submission Includes:

- ✓ Machine Learning Model
- ✓ Explainable AI Module
- ✓ Customer Segmentation
- ✓ Revenue Impact Analysis
- ✓ Recommendation Engine
- ✓ Power BI Dashboard
- ✓ Presentation Deck
- ✓ GitHub Repository

14. Final Product Vision

Our vision is to help businesses move from reactive customer management to proactive customer retention.

Rather than simply identifying churn, the platform empowers businesses to take the right action at the right time and retain valuable customers before they leave.